The defailed steps of Integration:

Show that $G_0(x) = \frac{b-a}{6}$ for closed newton cotes formula for n=2

$$G_0(x) = \int_a^b L_0(x)\,dx$$

$$= \int_a^b \frac{(x-m)(x-b)}{(a-m)(a-b)}\,dx$$

$$= \frac{1}{(a-m)(a-b)} \int_a^b (x-m)(x-b)\,dx$$

$$= \frac{1}{(a-\frac{a+b}{2})(a-b)} \int_a^b (x^2 - mx - bx + mb)\,dx$$

$$= \frac{1}{\left(\frac{a-b}{2}\right)(a-b)} \left[\frac{x^3}{3} - \frac{mx^2}{2} - \frac{bx^2}{2} + mbx\right]_a^b$$

$$= \frac{1}{\frac{(a-b)}{2}(a-b)} \left[\frac{b^3}{3} - \frac{mb^2}{2} - \frac{b^3}{2} + mb^2 - \frac{a^3}{3} + \frac{ma^2}{2} + \frac{ba^2}{2} - mba\right]$$

$$= \frac{2}{(a-b)^2} \left[\frac{2b^3 - 3mb^2 - 3b^3 + 6mb^2 - 2a^3 + 3ma^2 + 3ba^2 - 6mba}{6}\right]$$

$$= \frac{2}{(a-b)^2} \times \frac{1}{6} \times \left(-b^3 + 3mb^2 - 2a^3 + 3ma^2 + 3ba^2 - 6mba\right)$$

$$= \frac{2}{(a-b)^2} \frac{1}{6} \times \left(-b^3 + 3b^2\left(\frac{a+b}{2}\right) - 2a^3 + 3a^2\left(\frac{a+b}{2}\right) + 3a^2b - 6ab\left(\frac{a+b}{2}\right)\right]$$

$$= \frac{1}{3(a-b)^2} \left(-b^3 + \frac{3ab^2}{2} + \frac{3b^3}{2} - 2a^3 + \frac{3a^3}{2} + \frac{3a^2b}{2} + 3a^2b - \frac{6a^2b}{2} - \frac{6ab^2}{2}\right)$$

$$= \frac{1}{3(a-b)^2} \times \frac{-2b^3 + 3ab^2 + 3b^3 - 4a^3 + 3a^3 + 3a^2b + \cancel{6a^2b} - \cancel{6a^2b} - 6ab^2}{2}$$

$$= \frac{1}{3(a-b)^2} \times \frac{b^3 - 3ab^2 + 3a^2b - a^3}{2}$$

$$= \frac{(b-a)^3}{6(a-b)^2}$$

$$= \frac{\cancel{(b-a)^3}\,b-a}{6\cancel{(b-a)^2}} = \frac{b-a}{6} \qquad \left[(a-b)^2 = (b-a)^2\right]$$

$$G_1 = \frac{2}{3}(b-a)$$

Proof:

$$G_1 = \int_a^b \frac{(x-a)(x-b)}{(m-a)(m-b)}\,dx$$

$$= \frac{1}{(m-a)(m-b)} \int_a^b (x^2 - ax - bx + ab)\,dx$$

$$= \frac{1}{\left(\frac{a+b}{2}-a\right)\left(\frac{a+b}{2}-b\right)} \left[\frac{x^3}{3} - \frac{ax^2}{2} - \frac{bx^2}{2} + abx\right]_a^b$$

$$= \frac{1}{\left(\frac{b-a}{2}\right)\left(\frac{a-b}{2}\right)} \left[\frac{b^3}{3} - \frac{ab^2}{2} - \frac{b^3}{2} + ab^2 - \frac{a^3}{3} + \frac{a^3}{2} + \frac{a^2b}{2} - a^2b\right]$$

$$= \frac{1}{-\left(\frac{b-a}{2}\right)^2} \left[\frac{2b^3 - 3ab^2 - 3b^3 + 6ab^2 - 2a^3 + 3a^3 + 3a^2b - 6a^2b}{6}\right]$$

$$= \frac{1}{-\left(\frac{b-a}{2}\right)^2} \times \frac{-b^3 + 3b^2a - 3ba^2 + a^3}{6}$$

$$= \frac{4}{(b-a)^2} \times \frac{b^3 - 3b^2a + 3ba^2 - a^3}{6}$$

$$= \frac{\cancel{4}^{2}}{\cancel{6}\,3} \times \frac{(b-a)^3}{(b-a)^2} = \frac{2}{3}(b-a)$$

(Proved)